

Deterministic Optimization

Discrete Optimization:
Modeling w/ binary
variables

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Logical Relations

Modeling Logical Relations

Learning Objectives

- Observe some examples of modeling logical relations using mixed integer **linear** constraints

An investment example

- We would like to invest in 5 assets, labelled $\{1, 2, 3, 4, 5\}$
- We can only invest nonnegative amounts, and cannot invest more than a total of \$1000 dollars (budget)
- Let x_j denote the amount invested in asset j , then the constraints on our decisions are

$$0 \leq x_j, \quad \forall j = 1, \dots, 5, \quad \sum_{j=1}^5 x_j \leq 10000$$

Modeling Logical Relations using Binary variables

- We will model various logical constraints on our investment decisions
- We will make use of binary variables that indicate whether an investment as been made or not
- Introduce binary variable y_j which will take a value 1 if $x_j > 0$
- The constraints to force this are

$$0 \leq x_j \leq 1000y_j, \quad y_j \in \{0, 1\} \quad \forall j = 1, \dots, 5,$$

“OR” constraint

- We have to invest at least \$100 in asset 1 **or** asset 5
- Logically this constraint is: $(x_1 \geq 100) \vee (x_5 \geq 100)$
- Algebraically the constraint is:

$$x_1 \geq 100y_1, \quad x_5 \geq 100y_2, \quad y_1 + y_2 \geq 1$$

“XOR” Constraint

- We have to invest at least \$100 in asset 1 **or** asset 5 **but not both**
- Logically this constraint is: $(x_1 \geq 100) \vee (x_5 \geq 100)$
- Algebraically the constraint is:

$$x_1 \geq 100y_1, \quad x_5 \geq 100y_2, \quad y_1 + y_2 = 1$$

Implication

- **If** we invest \$100 or more in asset 1 **then** we must invest \$100 or more in asset 5
- Logically this constraint is: $(x_1 \geq 100) \Rightarrow (x_5 \geq 100)$
- Introduce a new binary variable z that gets set to 1 if $(x_1 \geq 100)$
- Algebraically the constraint can be written as:

$$z \in \{0, 1\}, \quad z \geq \frac{(x_1 - (100 - \epsilon))}{(900 + \epsilon)}, \quad 100z \leq x_5$$

where ϵ is a small positive number

Additional Examples

- If $x_1 > 0$ then $x_2 = 0$ or $x_3 = 0$. Constraint:

$$y_2 + y_3 \leq 2 - y_1$$

- If $x_1 > 0$ or $x_2 > 0$ then $x_5 = 0$. Constraint:

$$y_5 \leq 1 - y_1, \quad y_5 \leq 1 - y_2$$

Summary

- We saw some examples on how to model logical relations by mixed integer linear programming
- It is very important to make sure the resulting formulations are **linear**